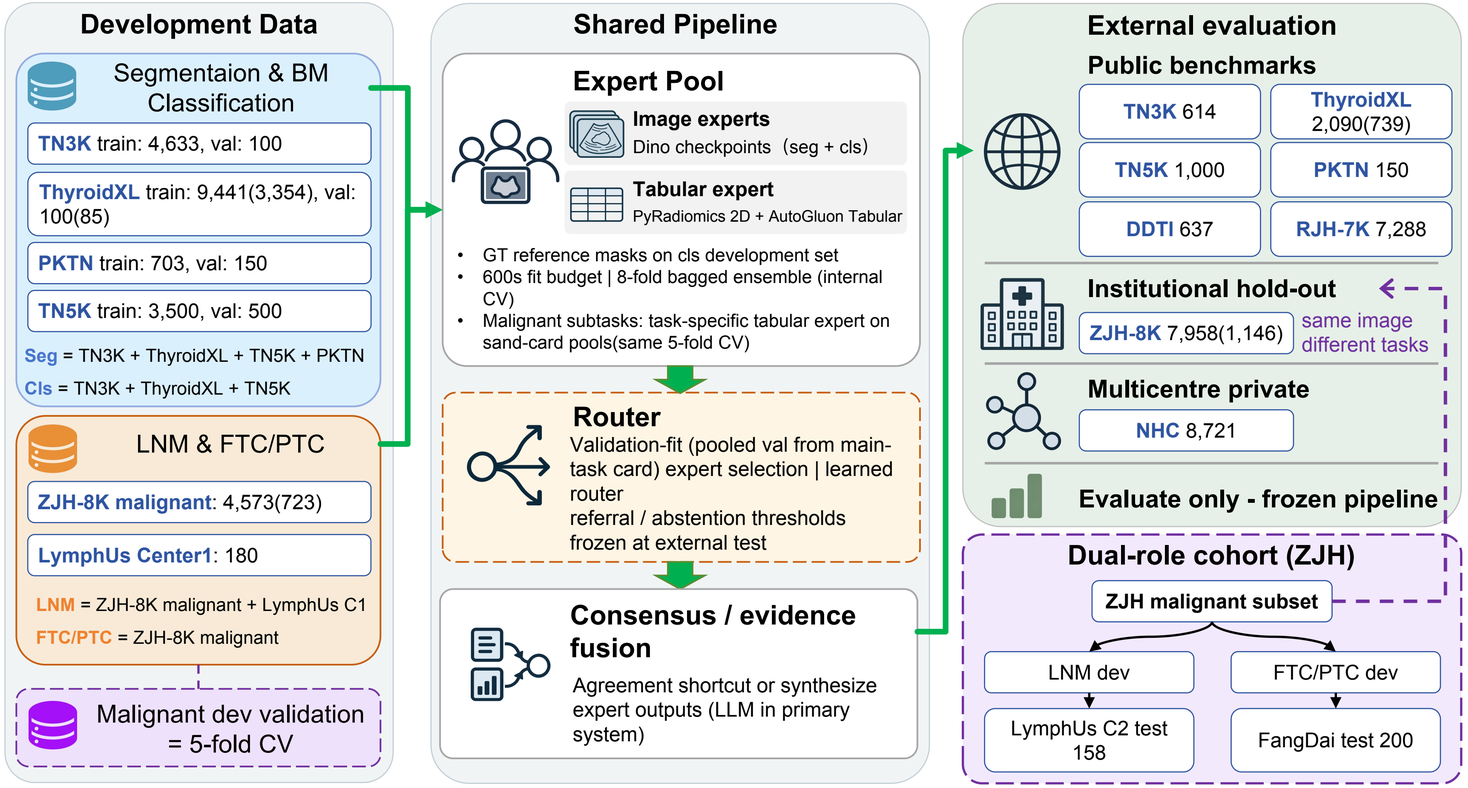

Development Data
Segmentaion & BM Classification
TN3K train: 4,633, val: 100
ThyroidXL train: 9,441(3,354), val: 100(85)
PKTN train: 703, val: 150
TN5K train: 3,500, val: 500
Seg = TN3K + ThyroidXL + TN5K + PKTN
Cls = TN3K + ThyroidXL + TN5K
LNM & FTC/PTC
ZJH-8K malignant: 4,573(723)
LymphUs Center1: 180
LNM = ZJH-8K malignant + LymphUs C1
FTC/PTC = ZJH-8K malignant
Malignant dev validation = 5-fold CV
Shared Pipeline
Expert Pool
Image experts
Dino checkpoints (seg + cls)
Tabular expert
PyRadiomics 2D + AutoGluon Tabular
GT reference masks on cls development set
600s fit budget | 8-fold bagged ensemble (internal CV)
Malignant subtasks: task-specific tabular expert on sand-card pools(same 5-fold CV)
Router
Validation-fit (pooled val from main-task card) expert selection | learned router
referral / abstention thresholds frozen at external test
Consensus / evidence fusion
Agreement shortcut or synthesize expert outputs (LLM in primary system)
External evaluation
Public benchmarks
TN3K 614
ThyroidXL 2,090(739)
TN5K 1,000
PKTN 150
DDTI 637
RJH-7K 7,288
Institutional hold-out
ZJH-8K 7,958(1,146)
same image different tasks
Multicentre private
NHC 8,721
Evaluate only - frozen pipeline
Dual-role cohort (ZJH)
ZJH malignant subset
LNM dev
FTC/PTC dev
LymphUs C2 test 158
FangDai test 200